Diego Arturo Alveal Vásquez

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Profile

Junior full-stack developer and Computer Science Engineer focused on AI integration (LLMs, agents, and RAG). Experienced in building web applications with modern technologies based on Clean Architecture, performing End-to-End tests and following good coding practices.

Experience

Universidad Católica de Temuco, CINAP Temuco, Chile
Undergraduate Thesis - AI-Assisted Scheduling Platform Jul 2025 - Jan 2026

- Built a scheduling platform for academic advising sessions with Next.js, FastAPI, and PostgreSQL following Clean Architecture, deployed with Docker.
- Implemented a LangGraph and LangChain conversational agent that books, reschedules, and cancels sessions through two MCP servers (FastMCP): one for Google Calendar and one for PostgreSQL.
- Migrated inference from Ollama to vLLM to serve Qwen3-4B on GPU, and added semantic search (RAG) with pgvector over institutional documents.
- Enabled scheduling through Telegram voice notes with Whisper transcription.
- Validated sub-2 s responses for multiple concurrent users on web, Telegram, and voice, with Redis caching.
- Used separate Git branches for development and testing before each production deployment.

Hospital San José de Victoria, IT Unit Victoria, Chile
Software Developer Intern Dec 2023 - Jan 2024

- Built a fingerprint identification web system (DigitalPersona U.are.U 4500) for supply staff to confirm purchase orders, replacing password-based identification.
- Implemented staff and fingerprint enrollment, missing-data checks, and tracking of who confirmed each order, using PHP, JavaScript, and a Docker service.
- Built a web app to register warehouse items by scanning barcodes from an industrial Android device, inserting and updating records in MySQL via AJAX.

Projects

Receipt Scanner | *Next.js, TypeScript, Tesseract.js, Convex* Nov 2026
project-ocr-receipts.vercel.app • github.com/DGVipe/project-ocr-receipts

- Built a web app that extracts the store, date, and total from receipt photos using in-browser OCR, without uploading the image to a server.
- Validated extracted data server-side and enforced per-user and per-IP rate limits through HMAC-hashed anonymous cookies, with no sign-up required.
- Added paginated lists, Excel export, and an English/Spanish interface.

Education

Universidad Católica de Temuco Temuco, Chile
Computer Science Engineering Graduated Oct 2026
Specialization in Advanced Topics in Software Development • GPA: 6.15 / 7.0

Skills

Programming: Python, TypeScript, JavaScript, SQL, Java, C, PHP
Frontend: Next.js, React, Tailwind CSS, shadcn/ui
Backend & Data: FastAPI, Flask, Laravel, Flask, Node.js, PostgreSQL, MySQL, SQL Server, Redis, Convex
AI & ML: LangGraph, LangChain, MCP (FastMCP), vLLM, Ollama, Whisper, RAG, Tesseract.js
Mobile & Tools: Flutter, React Native, Docker, Git, Vercel
Languages: Spanish (native); English (C1, advanced)